

# EduGenie

Google Gemini Powered Learning Assistant

## Phase 8 – Project Demonstration

Working Application Demonstration  
SmartBridge Generative AI Project

Document Date: September 23, 2026

*Demonstration of Completed Application*

# 1. Project Demonstration Overview

EduGenie is an AI-powered educational learning assistant designed to support students and learners in their educational journey. This demonstration presents the completed, fully functional application and illustrates how users interact with the system to access intelligent learning support.

**Purpose of Demonstration:** This phase showcases the working EduGenie application, demonstrating its user interface, five core learning functions, and how the system processes user input to generate AI-powered educational content.

**Key Points:** The demonstration focuses on the user perspective—how learners interact with the application, what options they have, and what types of educational results they receive from EduGenie.

# 2. Project Purpose

EduGenie addresses a fundamental challenge in modern education: providing accessible, personalized learning support that adapts to individual student needs. The application leverages Google Gemini's generative AI to deliver intelligent, contextual assistance across multiple learning scenarios.

**Learning Problems Addressed:**

- Students often need immediate answers to educational questions during study sessions
- Complex concepts can be difficult to understand from textbooks alone
- Self-assessment through quizzes is essential for learning reinforcement
- Large educational materials require summarization for efficient revision
- Personalized learning paths help learners progress systematically through topics

**How Generative AI Supports Learning:** EduGenie uses Google Gemini to generate contextual, educationally appropriate responses. Rather than providing rote answers, the AI creates explanations tailored to beginner learners, generates assessment quizzes, synthesizes information, and structures learning recommendations—all without requiring users to have technical AI knowledge.

# 3. Key Features Demonstrated

EduGenie implements five integrated learning functions. Each function accepts different types of user input and generates specialized educational output through AI processing.

| Learning Function | User Provides                            | System Generates                                  |
|-------------------|------------------------------------------|---------------------------------------------------|
| Ask a Question    | A specific question                      | Contextual, educational answer                    |
| Explain a Topic   | A concept or topic name                  | Simplified explanation with examples              |
| Generate Quiz     | Educational text + question count (1-10) | Multiple-choice quiz with 4 options per question  |
| Summarize Text    | An educational passage                   | Concise summary with key points                   |
| Learning Path     | Topic + level + available hours/week     | Structured learning plan with stages and timeline |

## **4. User Interface**

EduGenie presents a clean, modern interface designed for intuitive student interaction. The following screenshots show the actual application interface and its key components.

### **4.1 Main Interface and Task Selection**

The EduGenie interface features a prominent task selector dropdown containing all five learning functions. Users begin their learning session by selecting which type of assistance they need.

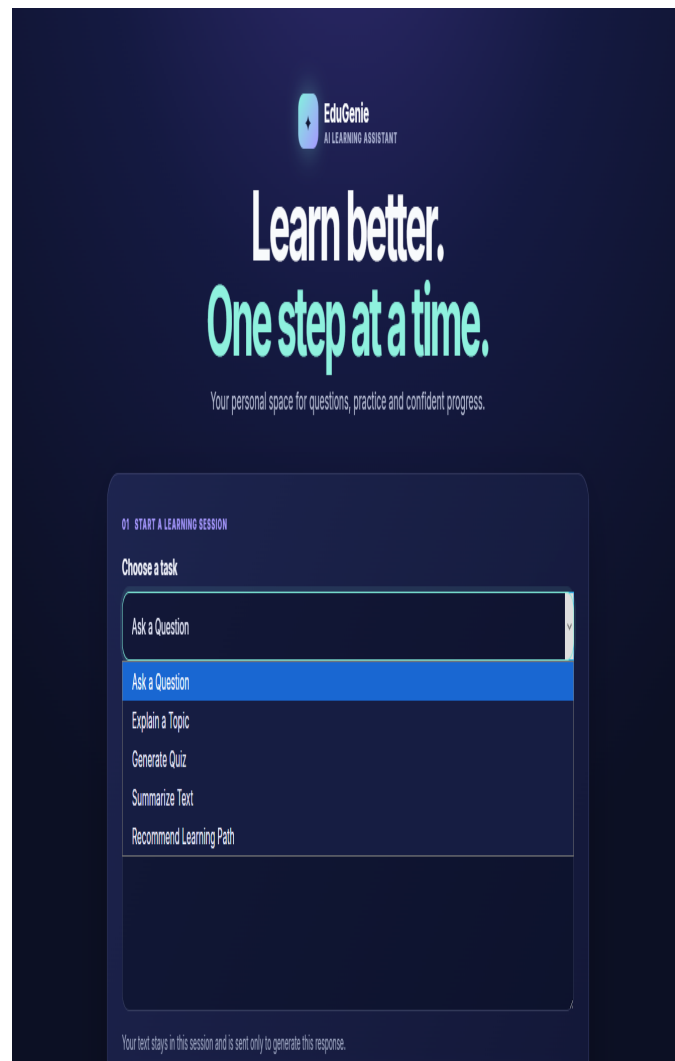


Figure 1: EduGenie Main Interface with Task Selection Dropdown

## 4.2 Ask a Question Interface

When 'Ask a Question' is selected, the interface updates to show a question input field with an example prompt. The Generate button submits the query to the backend, where it is processed by the AI system.

EduGenie

AI LEARNING ASSISTANT

Learn better.

One step at a time.

Your personal space for questions, practice and confident progress.

1. START A LEARNING SESSION

Choose a task

Explain a Topic

Topic

Explain The coding language Python

Your next steps in this session and is sent only to generate this response.

Generate

2. YOUR NEXT STEP

Your learning space

Complete

Alright, imagine you want to tell a computer to do something. Computers don't understand human languages like English directly. They need special instructions. That's where coding languages come in!

Let's talk about **Python**!

---

### 1. Simple Definition

Think of Python as a **special language** you use to talk to computers and give them instructions.

It's like learning a new language, but instead of talking to another person, you're talking to a machine. Python is known for being very clear and easy to read, almost like regular English, which makes it a great language for beginners!

---

### 2. How It Works

When you write instructions in Python, you're creating a "script" or "program." But the computer still doesn't understand Python directly. Here's what happens:

1. **You write your Python code:** You type out your instructions in a text file, like:

```
python
print("Hello, world!")
```

2. **The Python "Interpreter" steps in:** There's a special program on your computer called the "Python interpreter."

3. **It translates and executes:** When you tell your computer to run your Python script, the interpreter reads your code **"line by line"**. It translates each line into something the computer **"can"** understand and immediately performs that instruction.

So, it's like having a super-fast translator who reads your Python instructions one by one and tells the computer exactly what to do, right then and there!

---

### 3. Easy Example

Let's try a super simple example. We want the computer to say "Hello, world!"

**Your Python Code:**

```
python
# This is a comment - the computer ignores it, but it helps humans understand!
print("Hello, world!")
print(5 + 3)
```

**What happens when you run it:**

- The Python interpreter reads the first line (`# This is a comment...`) and ignores it.
- It reads `print("Hello, world!")`: The word `print` is a special instruction in Python that means "display this text on the screen."
- So, your computer screen will show: `Hello, world!`
- Then, it reads `print(5 + 3)`: It calculates `5 + 3` which is `8`.
- Then, it displays `8` on the screen.

**Output you would see:**

```
Hello, world!
8
```

See? It's like giving a direct command, and the computer follows it!

---

### 4. Key Points to Remember

-

EduGenie • FastAPI • Google Gemini

Figure 2: Ask a Question Interface with Input Field and Result Area

### 4.3 Generated Learning Output

After processing, the application displays the generated result in the 'Your Learning Space' section. The example shows an explanation of Python with structured sections including Definition, How It Works, Easy Example, and Key Points to Remember.

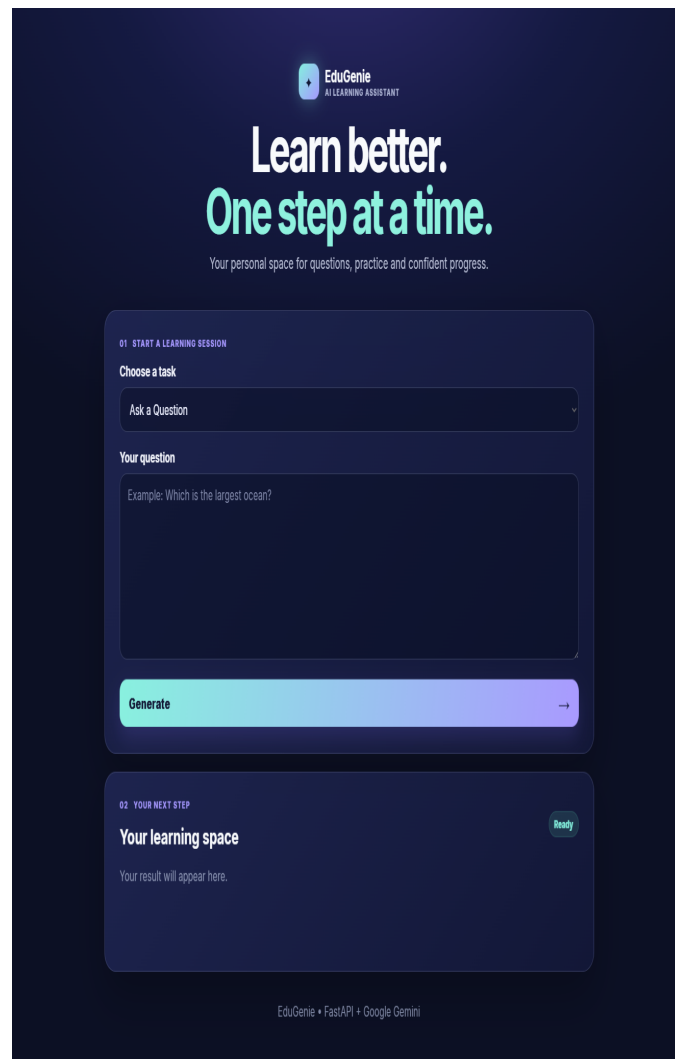


Figure 3: Generated Topic Explanation – Python Programming Explained for Beginners



## 5. Demonstration Workflow

The following steps outline the complete user interaction flow in EduGenie:

### Step 1 – Open EduGenie

User opens the application in a web browser at <http://127.0.0.1:8000>. The main interface loads with the hero section, task selector, and learning space.

### Step 2 – Select a Learning Function

User clicks the task dropdown and selects from: Ask a Question, Explain a Topic, Generate Quiz, Summarize Text, or Recommend Learning Path.

### Step 3 – Enter Required Input

User provides the appropriate input based on the selected task: a question, topic name, text passage, or learning preferences.

### Step 4 – Submit the Request

User clicks the green 'Generate' button to submit the request to the backend server.

### Step 5 – Backend Processing

The FastAPI backend receives the request, validates the input using Pydantic models, and routes it to the appropriate service module (QA, Explain, Quiz, Summary, or Learning Path).

### Step 6 – AI Processing

The service module constructs an optimized prompt and sends it to Google Gemini API. Gemini processes the prompt with configured parameters (temperature, token limits).

### Step 7 – Response Generation

Google Gemini generates contextual, educational content based on the prompt. For quizzes, responses are JSON-formatted and validated before returning.

### Step 8 – Backend Response

The backend receives Gemini's response, processes and formats it, then sends it back as JSON.

### Step 9 – Display Results

The frontend JavaScript receives the JSON response and displays it in the 'Your Learning Space' section. The user sees the generated educational content immediately.

## 6. Feature Demonstration

### 6.1 Ask a Question

**User Interaction:** User selects 'Ask a Question' from the dropdown, enters a specific educational question (e.g., 'What is photosynthesis?', 'How does gravity work?'), and clicks Generate.

**What the System Does:** The question is sent to Google Gemini with a prompt emphasizing accuracy, clarity, and educational appropriateness. Gemini generates a direct answer followed by simplified explanation and examples.

**Expected Result:** User receives a concise, educational answer suitable for a student, with explanations of important concepts and practical examples when relevant.

### 6.2 Explain a Topic

**User Interaction:** User selects 'Explain a Topic', enters a concept name (e.g., 'Python', 'Photosynthesis', 'Quantum Physics'), and clicks Generate. The screenshot demonstrates this feature with a Python

explanation.

**What the System Does:** The system sends the topic to Gemini with a specific prompt requesting: (1) Simple definition, (2) How it works, (3) Easy example, (4) Key points to remember. This structured prompt ensures beginner-friendly explanations.

**Expected Result:** User receives a well-structured explanation with clear sections, making complex topics accessible to beginners. As shown in Figure 3, the Python explanation includes simple definitions, working mechanisms, practical examples, and memorable key points.

## 6.3 Generate Quiz

**User Interaction:** User selects 'Generate Quiz', pastes an educational text or passage, selects the number of questions (1 to 10), and clicks Generate.

**What the System Does:** The backend sends the text and question count to Gemini with a prompt requesting JSON-formatted multiple-choice questions. Each question includes: question text, four options, correct answer, and explanation. The quiz module validates the JSON and ensures data quality before returning.

**Expected Result:** User receives a formatted quiz with the requested number of questions (1–10). Each question presents four plausible options, allowing learners to self-assess their understanding. Explanations help learners understand correct and incorrect options.

## 6.4 Summarize Text

**User Interaction:** User selects 'Summarize Text', pastes an educational passage or article, and clicks Generate.

**What the System Does:** The text is sent to Gemini with a prompt emphasizing retention of important facts, removal of repetition, use of simple language, and appropriate formatting with headings and bullet points where useful.

**Expected Result:** User receives a concise summary suitable for quick revision, preserving essential information while reducing the original text significantly. The summary uses simple language and may include bullet points or formatting for readability.

## 6.5 Recommend Learning Path

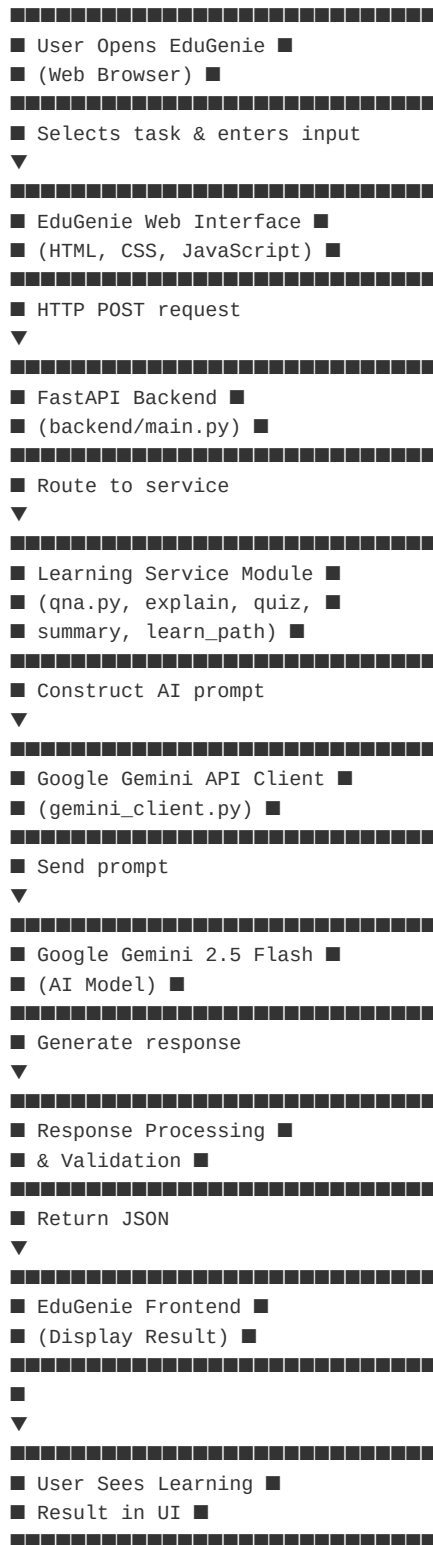
**User Interaction:** User selects 'Recommend Learning Path', enters a topic they want to learn, selects their current level (beginner, intermediate, advanced), specifies available study hours per week, and clicks Generate.

**What the System Does:** The system sends the topic, level, and study hours to Gemini with a comprehensive prompt requesting: learning stages, concepts to study at each stage, suggested timeline, practice ideas, resource types, a weekly routine, and milestones for each stage.

**Expected Result:** User receives a personalized learning plan structured by skill level and available time. The plan includes specific learning stages, daily/weekly schedules, practice suggestions, and checkpoints to measure progress.

## 7. AI and Backend Workflow

From the demonstration perspective, here is how user input becomes educational output:



## 8. Technology Used in the Demonstration

The EduGenie demonstration relies on the following technologies working together:

- **Python 3.10+** – Backend development language
- **FastAPI** – High-performance web framework handling HTTP requests
- **Uvicorn** – ASGI server running the FastAPI application
- **HTML5 / CSS3** – User interface markup and styling
- **JavaScript** – Frontend interaction and request handling
- **Jinja2** – Server-side template rendering
- **Google Gemini 2.5 Flash** – Generative AI model processing learning requests
- **google-genai SDK** – Python client for communicating with Gemini API
- **Pydantic** – Data validation for API requests

## 9. Application Execution

To run the EduGenie demonstration, follow these steps:

- 1. Environment Setup** – Create a Python virtual environment and install dependencies: `pip install -r requirements.txt`
- 2. Configure API Key** – Create a `.env` file in the project root and add your Google Gemini API key as: `GEMINI_API_KEY=your_key`
- 3. Start the Server** – Run: `python -m uvicorn backend.main:app --reload`
- 4. Open the Application** – Open your web browser and navigate to: `http://127.0.0.1:8000`
- 5. Access API Documentation** – View interactive API docs at: `http://127.0.0.1:8000/docs`

**Important Security Note:** Never commit your actual API key to version control. Always use environment variables to store sensitive credentials. The `.env` file should be added to `.gitignore`.

## 10. Demonstration Results

The provided screenshots and actual application execution demonstrate the following successful outcomes:

- ✓ EduGenie web interface loads successfully with professional, modern design
- ✓ User can access and interact with all five learning functions via the task dropdown
- ✓ Input fields accept user content and validate according to specified limits
- ✓ Generate button successfully submits requests to the backend
- ✓ FastAPI backend processes requests and routes them to appropriate service modules
- ✓ Google Gemini API integration successfully generates educational content
- ✓ Generated results display in real-time in the 'Your Learning Space' section
- ✓ Responses are formatted appropriately for each learning function type
- ✓ Quiz questions are properly formatted with options and explanations
- ✓ Error handling provides meaningful feedback when issues occur

## 11. Educational Benefits Demonstrated

EduGenie demonstrates practical educational benefits across multiple learning scenarios:

**Instant Question Answering:** Students receive immediate, accurate answers to educational questions without waiting for teacher response.

**Concept Simplification:** Complex topics are explained in beginner-friendly language with practical examples and key points.

**Self-Assessment:** Quiz generation from educational materials enables learners to test their understanding and identify knowledge gaps.

**Efficient Revision:** Text summarization condenses lengthy materials into concise summaries suitable for quick review before exams.

**Structured Learning:** Personalized learning paths provide clear progression from beginner to advanced levels with milestones and timelines.

**Accessible Learning Support:** Available 24/7 without subscription fees, EduGenie removes barriers to accessing quality educational assistance.

## 12. Demonstration Limitations

The current demonstration and application have the following limitations that should be noted:

- **No User Accounts:** The current version does not support user registration or persistent learning history.
- **Session-Based Only:** User interactions are not saved; each session starts fresh with no history retained.
- **API Dependency:** The application requires internet connectivity and Google Gemini API availability.
- **Input Limits:** Questions are limited to 10,000 characters; text passages to 20,000 characters.
- **Quiz Format:** Quiz generation requires Gemini to return valid JSON; complex topics may occasionally fail.
- **Response Accuracy:** AI-generated content may occasionally contain inaccuracies or require verification.
- **English Only:** Currently optimized for English language input and output.
- **No Offline Mode:** Application requires live API connectivity; offline use is not supported.
- **Browser Only:** Currently a web application; mobile native apps are not available.

## 13. Future Enhancement Opportunities

The following enhancements are planned to extend EduGenie's capabilities:

- User accounts with secure authentication for persistent learning history
- Database integration for saving sessions, quiz results, and learning progress
- Mobile applications (iOS/Android) for learning on-the-go
- Multilingual support for global learner accessibility
- Progress tracking and analytics showing learning patterns and improvements

- Adaptive learning paths that adjust based on quiz performance
- Voice interaction for hands-free learning
- Teacher/educator dashboards for classroom integration
- Learning Management System (LMS) integration
- Gamification features including badges, points, and leaderboards
- Image and PDF document support for explaining diagrams
- Group study features for collaborative learning

## 14. Demo Video

A complete demonstration video showing the EduGenie application in action is available at the following link:

**Demo Video Link:**

[\[Google Drive Link - To Be Added\]](#)

Access: Anyone with the link

The video demonstrates all five learning functions, shows response generation in real-time, and explains the educational value of each feature.

## 15. Demonstration Checklist

The following checklist confirms the elements demonstrated in the completed EduGenie project:

- Project name (EduGenie) introduced
- Project purpose explained (AI-powered learning assistant)
- EduGenie application opened in web browser
- Main user interface with hero section demonstrated
- Task selection dropdown shown with all five options
- Ask a Question feature demonstrated
- Explain a Topic feature demonstrated with Python example
- Generate Quiz feature demonstrated
- Summarize Text feature demonstrated
- Learning Path Recommendation feature demonstrated
- Generated outputs displayed in 'Your Learning Space'
- Real-time response generation shown
- Backend processing explained
- Google Gemini AI integration demonstrated
- Application responsiveness verified
- Error handling shown (if applicable)

## 16. Conclusion

The EduGenie Phase 8 demonstration presents a fully functional, AI-powered learning assistant that successfully integrates Google Gemini with an intuitive web interface. The demonstration showcases all five core learning functions: question answering, topic explanation, quiz generation, text summarization, and personalized learning path recommendations.

Through the provided screenshots and application execution, users can observe how learners interact with EduGenie, the types of input the system accepts, and the quality of educational output generated by the AI. The application demonstrates practical benefits for students seeking accessible, immediate learning support.

EduGenie successfully fulfills the objectives of Phase 8 (Project Demonstration) by presenting a working application ready for real-world use. The demonstration proves that generative AI can effectively support educational goals through thoughtful system design, appropriate prompt engineering, and user-centered interface development.

This concludes the SmartBridge development lifecycle for EduGenie, from initial ideation through final demonstration. The project is complete, functional, and ready for future enhancement and scaling.



*Document Generated: September 23, 2026  
EduGenie Phase 8 – Project Demonstration  
SmartBridge Development Lifecycle – Completed*